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Influence of the ocean loading effect on GPS derived precipitable water vapor

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Abstract. For meteorology the precipitable water vapor becomes a more and more important product in the data analysis of the Global Positioning System (GPS). It is derived from the wet part of the tropospheric refraction obtained as zenith path delay in the data analysis. However, the estimates of the zenith path delay and of the station height are strongly correlated.

The ocean loading effect has a periodic influence on station heights with major frequencies in the diurnal and semidiurnal band and with amplitudes up to several centimeters. If one ignores this effect it affects the estimates of the zenith path delay and thus the precipitable water vapor. This will be demonstrated by some examples.

1. Introduction

To compute the precipitable water vapor from GPS data the zenith path delay has to be estimated in the data analysis. In any elevation, the delay is caused by the influence of the tropospheric refraction on the observations. A so-called mapping function has then to be used to relate the effect of the zenith delay to a delay at a lower elevation. The delay can be divided into two parts, a hydrostatic and a (remaining) wet part [*Davis et al.,* 1985]. The wet part is given by the temperature and the water vapor.

Usually, when estimating water vapor with GPS, the parameters temperature and air pressure are assumed to be known from ground measurements in the surroundings of the GPS antenna [*Bar-Sever,* 1996]. That means, that any error in the estimated zenith path delay can be found in the derived precipitable water vapor with its full influence. The formalism is shown, for instance, in *Bevis et al.,* [1994]. It ends up with a simple rule of thumb: a zenith path delay of 1 mm corresponds to about 0.15 mm of precipitable water vapor for mean atmospheric conditions.

The estimates of the station height and the zenith path delay are strongly correlated [*Rothacher and Beutler,* 1998]. Therefore, one has to model all effects on the station height very carefully to get reliable zenith path delays and to avoid systematic errors [*Fang et al.,* 1998]. One of the model components, which may be important in this context, is the vertical crustal deformation induced by ocean tides.

Ocean loading leads to periodic variations of station heights. *Scherneck* [1993] has shown that a global error of 5 cm for the ocean tides leads to ocean loading uncertainties for the vertical crustal deformation below the millimeter level even for coastal stations. Today the global accuracy of modern ocean tide models is generally considered to be in the few centimeter level [*Shum et al.,* 1997]. Therefore, the ocean loading effect can be assumed as error-free with respect to GPS derived zenith path delays – even in polar regions, where ocean tide information are less reliable. We decided to use the FES95.2.1 [*LeProvost et al.,* 1998] model for the loading computations.

2. The GPS Data Analysis

2.1. Software, Analysis Strategy and Data

For the numerical investigations presented here the Bernese GPS Software, version 4.0, was used [*Beutler et al.,* 1996]. The software performs a double difference analysis for baseline computations. Hereby, zenith path delay and mapping function are taken from the Saastamoinen model [*Saastamoinen,* 1973]. In the standard version tidal loading corrections are not applied. In an unconstrained parameter estimation procedure as carried out in this study station coordinates, ambiguities and the zenith path delay (updated twelve times a day) are determined. The zenith path delays are estimated as relative values for baseline lengths of up to 150 km and as absolute values for longer baselines, respectively. However, in this paper the differences of the estimated values are shown and discussed also for baselines longer than 150 km. In all cases an elevation mask of 10^o has been applied.

In the simulation study relative effects were investigated for a single baseline of 100 km length, which is located at a latitude of 60^o south.

Real data were taken from the region of the Antarctic Peninsula. The geographical distribution of the stations is shown in figure 1. These stations are part of the network of SCAR GPS Epoch Campaigns, organized by the Scientific Committee on Antarctic Research Working

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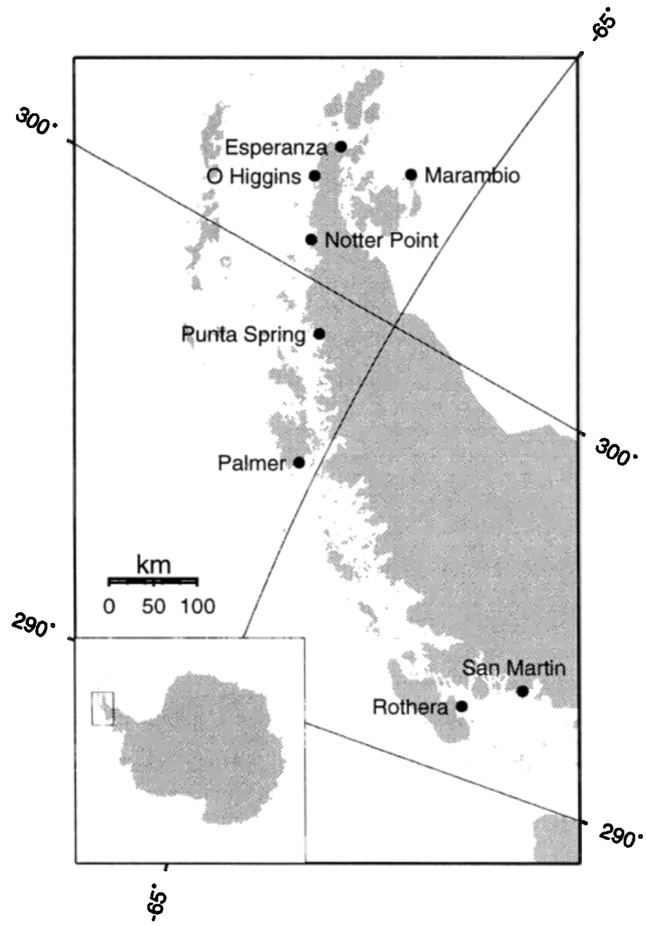

Figure 1. Map with the location of the GPS stations in the region of the Antarctic Peninsula

Group on Geodesy and Geographic Information [*Dietrich et al.,* 1998]. The stations were occupied with Trimble 4000SSi receivers. From the 1998 campaign three weeks of data were available, with a sampling rate of 15 seconds.

2.2. Simulation Study

In the simulation study synthetic observations were generated. The effects of the tropospheric refraction were neglected. Therefore, the estimated zenith path delay can expected to be zero. A relative ocean loading effect was simulated with one diurnal and one semidiurnal frequency and an amplitude of 8 mm each. For the difference of the ocean loading effect this order of magnitude can be considered as typical for a baseline of about 100 km, where one station is situated at the coast and the other one inland.

In the estimation procedure the ocean loading effect is ignored, while the coordinates and the zenith path delay are solved. For the latter every two hours new parameters are determined. The result is shown in figure 2. The values for the tropospheric zenith path delay follow the simulated ocean loading function very well. Because tropospheric effects were not introduced into the data simulation, the amplitude of the zenith delay curve is a result of the ignored loading effect only. Introducing the assumed ocean loading function into the model the computed zenith path delay becomes less than 0.2 mm.

Table 1. Estimated scaling factors for the relative ocean loading effect derived from the FES95.2.1 model; all baselines from Marambio.

Baseline to	Baseline length	Magnitude of relative ocean loading	Scaling factor
Esperanza	96 km	10 mm	1.13
O'Higgins	120 km	10 mm	1.12
Notter Point	140 km	10 mm	1.02
Punta Spring	213 km	13 mm	1.05
Palmer	360 km	21 mm	1.11
San Martin	638 km	30 mm	1.04
Rothera	640 km	31 mm	1.00

2.3. Validation of the Ocean Loading Model with GPS Data

At first the validity of the ocean loading model in the GPS data analysis has to be checked. For this purpose, the ocean loading effect is introduced into the model and scaled by a factor, which is added as a further unknown parameter into the standard estimation procedure of the Bernese GPS Software. If the model of the applied ocean loading correction is fully true this scale factor has to be one, and it becomes zero in the case that the effect of the model is completely missing in the data.

Table 1 shows the estimates of these scale factors for baselines of selected stations in the region of the Antarctic Peninsula. The resulting values are in the range between 1.00 and 1.13. A rescaling of the relative ocean loading effects by the estimated scale factors corresponds to an additional relative vertical displacement of 2 mm in the maximum case. Hence, the validity

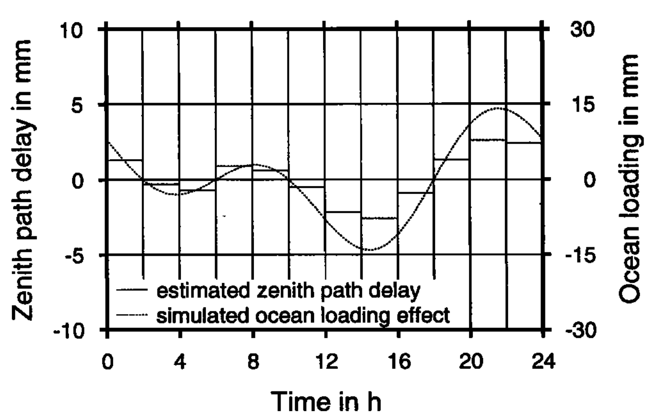

Figure 2. Zenith path delay estimates when ocean loading deformation is ignored in the analysis (simulated GPS data: ocean loading signal introduced, tropospheric delay excluded)

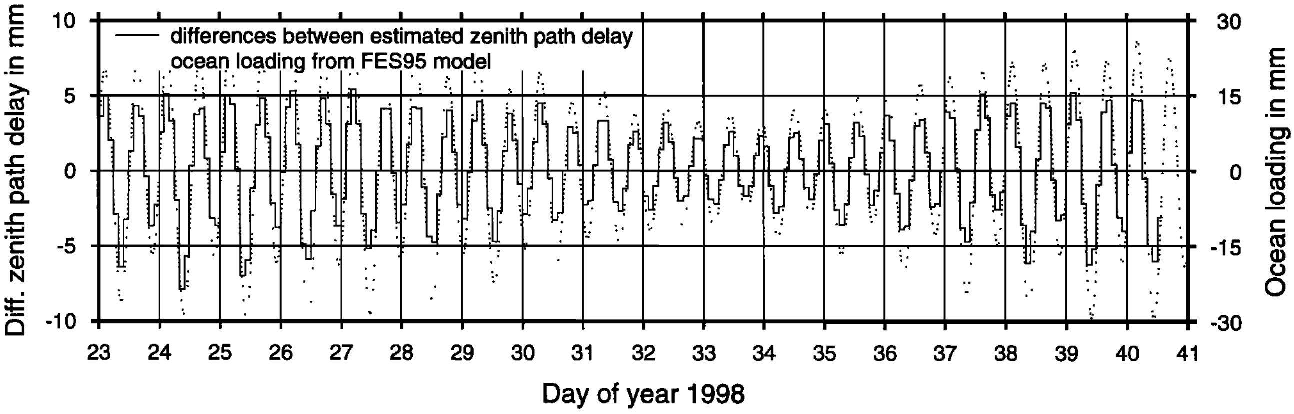

Figure 3. Differences between the estimated zenith path delays from the GPS solutions for the baseline from Marambio to San Martin (length 638 km) introducing a model of ocean loading deformation based on FES95.2.1 and without such corrections

of the ocean loading effect derived from the FES95.2.1 ocean tide model is demonstrated.

2.4. Influence of the Loading Effect on Zenith Path Delays

To show the influence of the tidal loading on the zenith path delay two GPS solutions were performed: one solution where the ocean loading effect was introduced, and another where it was completely ignored. The other analysis options were kept identically.

Figure 3 shows an example of the differences between the estimated zenith path delays of these two solutions. The chosen baseline between the stations Marambio and San Martin has a length of more than 600 km. For Marambio the effect of the vertical crustal deformation can reach up to 70 mm, while for San Martin it is up to 44 mm. The maximum difference of the loading effect, for these two stations is 30 mm, as shown in figure 3.

Ignoring this effect will bias the estimated relative zenith path delay by up to 8 mm, which – following the rule of thumb mentioned in the introduction – is equivalent to about 1.2 mm precipitable water vapor. The mean ratio between the zenith path delay and the vertical crustal deformation becomes 0.21 with a standard deviation of 0.01. To conclude, one can find more than 20% of the vertical loading effect in the zenith path delay estimates. This result confirms the conclusion of the simulation study.

The correlation between the vertical deformation induced by the ocean tides and the differences of the estimated zenith path delays with and without corrections for the ocean loading effect comes up with 0.99. Nevertheless, this may confirm that the estimation of the zenith path delay from GPS data is a stable procedure, even for such a long baseline.

The solution presented here was computed using an elevation mask of 10°. Excluding all observations up to an elevation angle of 20° the ratio between the zenith path delay and the crustal deformation becomes 0.35±0.03. That means, that about one third of the vertical crustal deformation can be found in the estimated zenith path delays when ignoring the ocean loading effect. For the baseline from Marambio to San Martin with a differential loading signal up to 30 mm the effect is more than 10 mm for the relative zenith path delay. This value corresponds to about 1.5 mm precipitable water vapor.

For other baselines in the region of the Antarctic Peninsula the resulting ratios between the GPS derived zenith path delays and the crustal deformation induced by ocean tides are shown in table 2. These values confirm the result shown for the baseline Marambio – San Martin in figure 3: Using an elevation mask of 10 degree about 22% of the ocean loading effect can be found in the estimated zenith path delays. In general, the correlation between the ocean loading signal and the zenith path delay is above 0.9 in all cases (table 2, last column).

3. Summary

This study focuses on the influence of the ocean loading effect on the zenith path delay estimates from GPS data. It could be shown that about 20% of the effect of

Table 2. Statistics of the differences of the estimated zenith path delays with respectively without using the corrections for the ocean loading effect in relation to the ocean loading effect

Baselines from	to	Ratio	Correlation
Marambio	Esperanza	0.21 ± 0.01	0.99
Marambio	O'Higgins	0.22 ± 0.02	0.99
Marambio	Notter Point	0.21 ± 0.01	0.99
Marambio	Punta Spring	0.26 ± 0.02	0.96
Marambio	Palmer	0.21 ± 0.01	0.99
Marambio	San Martin	0.21 ± 0.01	0.99
Marambio	Rothera	0.19 ± 0.03	0.92

umodeled vertical crustal deformations can be found in zenith path delay estimates for GPS solutions with an elevation mask of 10°. A numerical test using GPS data from Antarctica shows that the effect of the neglected ocean tidal loading may change the estimated relative zenith path delay by up to 10 mm.

A comparison of zenith path delay estimates from different analysis centers of the International GPS Service for Geodynamics (IGS) was performed by *Gendt* [1998]. A consistency at the 4 to 5 mm level was concluded. In many cases a standard deviation of ±3 mm is reached. If one takes these values as the present accuracy level for zenith path delay estimates from GPS, our study confirms that the ocean loading effect should to be taken into account.

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References

Bar-Sever, Y. E., Strategies for Near Real Time Estimation of Precipitable Water Vapor, paper presented at the IGS 1996 Analysis Center Workshop, Silver Spring, Maryland U.S.A, Mar. 19 to 21, 1996.

Bevis, M., S. Businger, S. Chiswell, T. A. Herring, R. A. Anthes, C. Rocken, and R. H. Ware, GPS Meteorology: Mapping Zenith Wet Delays onto Precipitable Water, *Journal of Applied Meteorology 33,* 379-386, 1994.

Beutler, G., E. Brockmann, S. Fankhauser, W. Gurtner, J. Johnson, L. Mervart, M. Rothacher, S. Schaer, T. Springer, and R. Weber, Bernese GPS Software Version 4.0, Documentation, 1996.

Davis, J.L., T. A. Herring, I. I. Shapiro, A. E. E. Rogers, and G. Elgered, Atmospheric Modeling Errors on Estimates of Baseline Length, *Radio Science 20,* 1593–1607, 1985.

Dietrich, R., R. Dach, J. Perlt, H.-W. Schenke, T. Schöne, M. Pohl, G. Soltau, G. Engelhardt, H.-W. Mikolaiski, G. Seeber, F. Menge, W. Niemeier, H. Salbach, K. Lindner, H.-J. Kutterer, and M. Mayer, The SCAR GPS Campaigns: Accurate Geodetic Reference in Antarctica, in Forsberg, R., M. Feissel, and R. Dietrich, editors, *IAG Scientific Assembly, Rio de Janeiro, Sept. 3-9, 1997,* Springer Series: IAG Symposia, Vol. 119, Springer Heidelberg, 474-479 1998.

Fang, P., M. Bevis, Y. Bock, S. Gutman, and D. Wolfe, GPS meteorology: Reducing systematic errors in geodetic estimates for zenith delay, *Geophys. Res. Lett., 25*(9), 3583–3586, 1998.

Gendt, G., IGS Combination of Tropospheric Estimates – Experience from Pilot Experiment, the IGS 1998 Analysis Center Workshop, Darmstadt, Germany, Feb. 9 to 11, 1998.

Langley, R. B., Propagation of the GPS Signals, in Teunissen, P. J. G. and Kleusberg, A., eds., *GPS for Geodesy,* 2nd Edition, Springer Verlag, Berlin - Heidelberg, 1998, pp. 111-150.

LeProvost, C., L. Lyard, J. Kolines, M. Genco and F. Rabilloud, A hydrodynamic ocean tide model improved by assimilating a satellite altimeter derived dataset, *J. Geophys. Res., 103*(C3), 5513–5529, 1998.

Rothacher, M., and G. Beutler, The Role of GPS in the Study of Global Change, *Chemistry and Physics of the Earth, 2,* 1998.

Saastamoinen, I. I., Contribution to the theory of atmospheric refraction, *Bulletin Géodésique, 107,* 13-34, 1973.

Scherneck, H.-G., Ocean tide loading: propagation of errors from ocean tide into loading coefficient, *manuscripta geodetica 18*, 59–71, 1993.

Shum, C., P. Woodworth, O. Andersen, G. Egbert, O. Francis, C. King, S. Klosko, C. LeProvost, X. Li, J. Molines, M. Parke, R. Ray, M. Schlax, D. Stammer, C. Tierney, P. Vincent, and C. Wunsch, Accuracy Assessment of Recent Ocean Tide Models, *J. Geophys. Res., 102*(C11), 25173–25194, 1997.

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